



MODULE NAME:	MODULE CODE:
SYSTEM ANALYSIS & DESIGN	SAND6211/d/p
SYSTEM ANALYSIS & DESIGN	SAND6221/d

ASSESSMENT TYPE:	TAKE-HOME ASSESSMENT (PAPER ONLY)
TOTAL MARK ALLOCATION:	120 MARKS
TOTAL TIME:	This assessment should take you 2 Hours to complete, however you have 21 Hours (midnight to 9PM on the same day) to submit. This additional time has been allocated to allow for the download, completion and upload of your submission.

By submitting this assessment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. Please **adhere to all instructions**. These instructions are different from what is normally present, so take time to go through these carefully.
2. **Independent work is required**. Students are not allowed to work together on this assessment. Any contraventions of this will be handled as per disciplinary procedures in The IIE policy.
3. **No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks.**
4. All work must be adequately and correctly referenced.
5. You should paraphrase (use your own words) the concepts that you are referencing, rather than quoting directly.
6. This is an open-book assessment.
7. Assessments must be typed unless otherwise specified.
8. **Ensure that you save a copy of your responses.**
 - a. Complete your responses MS WORD
 - b. Create a folder in the following format: **StudentNumber_ModuleCode_Exam** and save all your work in this folder. Once you have completed the assessment, upload your document under the **submission link** in the correct module in Learn.

Additional instructions:

- Answer All Questions.

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Hence, The IIE considers it essential to develop the referencing skills of our students in our commitment to achieve high academic standards. Part of achieving these high standards is referencing in a way that is consistent, technically correct and congruent. This is not plagiarism, which is handled differently.

Poor quality formatting in your referencing will result in a penalty **of a maximum of ten percent being deducted from the percentage awarded**, according to the following guidelines. Please note, however, that **evidence of plagiarism in the form of copied or uncited work (not referenced), absent reference lists, or exceptionally poor referencing, may result in action being taken in accordance with The IIE's Intellectual Integrity Policy (0023).**

Markers are required to provide feedback to students by indicating **(circling/underlining) the information that best describes the student's work.**

Minor technical referencing errors: 5% deduction from the overall percentage – the student's work contains **five or more errors** listed in the minor errors column in the table below.

Major technical referencing errors: 10% deduction from the overall percentage – the student's work contains **five or more errors** listed in the major errors column in the table below.

If both minor and major errors are indicated, then 10% only (and not 5% or 15%) is deducted from the overall percentage. The examples provided below are not exhaustive but are provided to illustrate the error

<u>Required:</u> Technically correct referencing style	<u>Minor errors</u> in technical correctness of referencing style Deduct 5% from percentage awarded	<u>Major errors</u> in technical correctness of referencing style Deduct 10% from percentage awarded
<u>Consistency</u> The same referencing format has been used for all in-text references and in the bibliography/reference list.	Minor inconsistencies. - The referencing style is generally consistent, but there are one or two changes in the format of in-text referencing and/or in the bibliography. - For example, page numbers for direct quotes (in-text) have been provided for one source, but not in another instance. Two book chapters (bibliography) have been referenced in the bibliography in two different formats.	Major inconsistencies. - Poor and inconsistent referencing style used in-text and/or in the bibliography/ reference list. - Multiple formats for the same type of referencing have been used. - For example, the format for direct quotes (in-text) and/or book chapters (bibliography/ reference list) is different across multiple instances.
<u>Technical correctness</u> - Referencing format is technically correct throughout the submission. - Position of the reference: a reference is directly associated with every concept or idea. - For example, quotation marks, page numbers, years, etc. are applied correctly, sources in the bibliography/reference list are correctly presented.	Generally, technically correct with some minor errors. - The correct referencing format has been consistently used, but there are one or two errors. - Concepts and ideas are typically referenced, but a reference is missing from one small section of the work. - Position of the references: references are only given at the beginning or end of every paragraph. - For example, the student has incorrectly presented direct quotes (in-text) and/or book chapters (bibliography/reference list).	Technically incorrect. - The referencing format is incorrect. - Concepts and ideas are typically referenced, but a reference is missing from small sections of the work. - Position of the references: references are only given at the beginning or end of large sections of work. - For example, incorrect author information is provided, no year of publication is provided, quotation marks and/or page numbers for direct quotes missing, page numbers are provided for paraphrased material, the incorrect punctuation is used (in-text); the bibliography/reference list is not in alphabetical order, the incorrect format for a book chapter/journal article is used, information is missing e.g. no place of publication had been provided (bibliography); repeated sources on the reference list.
Congruence between in-text referencing and bibliography/ reference list All sources are accurately reflected and are all accurately included in the bibliography/ reference list.	Generally, congruence between the in-text referencing and the bibliography/ reference list with one or two errors. - There is largely a match between the sources presented in-text and the bibliography. - For example, a source appears in the text, but not in the bibliography/ reference list or vice versa.	A lack of congruence between the in-text referencing and the bibliography. - No relationship/several incongruencies between the in-text referencing and the bibliography/reference list. - For example, sources are included in-text, but not in the bibliography and vice versa, a link, rather than the actual reference is provided in the bibliography.
In summary: the recording of references is accurate and complete.	In summary, at least 80% of the sources are correctly reflected and included in a reference list.	In summary, at least 60% of the sources are incorrectly reflected and/or not included in reference list.

Overall Feedback about the consistency, technical correctness and congruence between in-text referencing and bibliography:

Question 1		(Marks: 30)
Answer all of the questions below.		
Q.1.1	Explain the need for a System Development Process in Software Development Projects.	(2)
Q.1.2	Define what a Object Oriented Design is in Systems Analysis and Design. Please use your own words.	(2)
Q.1.3	Distinguish between internal and external stakeholders in systems analysis. Please use your own words.	(2)
Q.1.4	Distinguish between brief use case description and fully developed use case description . Please use your own words.	(4)
Q.1.5	Distinguish between system analysis and system design , and discuss the importance of the two in Software Development.	(4)
Q.1.6	Briefly explain what predictive approach to SDLC is.	(2)
Q.1.7	Define user-centered design in designing user interfaces and list three of its important principles.	(5)
Q.1.8	Define what usability is in user interface design.	(1)
Q.1.9	Discuss two things you would take into consideration when designing the interface for both Web and Mobile.	(4)
Q.1.10	Distinguish between Database and Database Management System (DBMS). List two real-life examples of a DBMS . Please use your own words.	(4)

Question 2**(Marks: 30)**

The questions below relate to the following case study:

System: Online University Platform

Pune Institute of Technology is an Online University based in Pune, India. The university's main goal is to address Technology Skills shortages globally. The University offers Information Technology courses to students all over the world through distance learning. All the prospective student need is a computer and stable internet connection. As part of its global outreach strategy, the online university has to implement robust online platform to simplify enrolment, communication, billing and distance learning in general.

You are part of the Project Team that is appointed to spearhead the development of the online platform. As a Software Development Consultant, you are also required to document requirements of the online platform and design the overall system applying good design principles. You are also required to study agile project management approach as its going to be applied in the software development project. The Project Team will consist of a Project Manager, Product Owner, Scrum Master and Development Team (Software Development Consultants with different specialities).

As part of the business processes the system has to cater for, the Online Technical Assistant will load online tasks, review the tasks, maintain the versioning of the tasks, assign online tasks to students, assess and moderate the tasks, aggregates the results of the completed tasks, and captures them. The Course Facilitators will oversee and supervise the work of the Online Technical Assistants. Course Facilitators report to the Head of Department (HoD). The Course Facilitator's duty in the workflow will be to moderate the results after they are captured by the Online Technical Assistant, approves the results, sign them off and submits them for capturing. The results will end up in the HoD's Office. The HoD receives the results reports, moderates the reports, selects best performer and issues the final approval. The office of the HoD has a full-time Office Manager. The Office Manager supports the Online University Platform by constantly updating students academic records, unblocking blocked courses on behalf of students and doing general updates to course records. It is also the duty of the Office Manager to enroll new students on the Online University Platform.

Results are captured per Course, a Course belongs to a Department and a Department belongs to a Faculty within the University. Students can enroll for more than one Course and Online Technical Assistants are allowed to facilitate many Courses during the course of the year. The Online University Platform should also have a functionality for students to register, check course costs, download proof of registration, update contact details, and download statement of account compiled by finance. All the above functionalities will be delivered in the first phase of the Project.

Quality is valued at the University and no project is approved if this aspect cannot be demonstrated. Before deploying the system in Production, the project team will embark on a data take-on exercise because an operational system requires a fully populated database to support ongoing processing. As stated, that Quality is valued at the University, the development team is forced to apply strict rules around Unit Testing and Integration Testing. Performance or stress testing report will form part of the approval process before the system is deployed into production. To save money on database licencing the University mostly uses open source software and relational database management system (DBMS). MySQL is a free and open-source DBMS to be used during the Project. All the databases on MySQL including the back-ups should be encrypted.

After the deployment of the system in production, the project team will embark on the second phase of the Project. Other functionalities of the Online University Platform that will be catered for in the second phase, are the ability for Students to lodge disputes, view results, rate the course content, like the content, comment on the content and chat with Online Technical Assistants. The Online University Platform should also be accessible on a Mobile Device.

Q.2.1	Develop a UML activity diagram to model the activities between the Online Technical Assistant, the Course Facilitator and the HoD.	(30)
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Question 3		(Marks: 40)
The questions below are still based on the case study in Question 2.		
Q.3.1	The project team will embark on a data take-on exercise because an operational system requires a fully populated database to support ongoing processing. Discuss how a DBMS interface works to extract data from a database.	(3)
Q.3.2	In today's interconnected world, systems need reliable access control systems to keep the data secure. Define the three elements that access control systems rely on.	(6)
Q.3.3	Agile project management approach is going to be applied in the software development project. Define agile development and list its four basic values as enshrined in the "Manifesto for Agile Software Development".	(6)
Q.3.4	Identify use cases in relation to the Student's interaction with the Online University Platform and develop a use case diagram.	(21)
Q.3.5	We need good design principles in order to design high-quality software. Define what separation of responsibilities and protection from variations is in system design.	(4)

Question 4		(Marks:20)
The questions below are still based on the case study in Question 2.		
Q.4.1	The development team is forced to apply strict rules around Unit Testing and Integration Testing.	(4)

	Discuss the need for Testing in Software Development. Please use your own words.	
Q.4.2	Performance or stress testing report will form part of the approval process before the system is deployed into production. Define time-based performance criteria when conducting performance or stress testing.	(4)
Q.4.3	Quality is valued at the University and no project is approved if this aspect cannot be demonstrated. Define a type of system test that allows business stakeholders to check system functionality against user requirements.	(2)
Q.4.4	As stated in the case study, all the databases on Postgres including the back-ups should be encrypted. Discuss the importance of encryption and distinguish between encryption and decryption in computer security.	(6)
Q.4.5	You are going to need to put few architectures in place to meet all the requirements for the Online University Platform. Distinguish between Technology and Application Architecture.	(4)

END OF PAPER